Lesson plan · for the teacher

Inside a language model

AI · Level 3 · Lesson 4

Time
40 min

Materials
Board and coloured chalk; Three sentences where “it” could mean two things; A page from your textbook (for the grounding demonstration)

Learning goals

- Describe the loop: tokens become numbers, attention gathers context, the model scores every possible next token, picks one and repeats.
- Explain the training stages: pretraining on huge amounts of text, then fine-tuning with examples and human feedback.
- Explain why a language model can be confidently wrong, and how grounding and checking help.

Activity: Who is “it”?

1. Warm-up (5 min): Write “The bus didn’t stop at the school because it was…” on the board. Pupils call out endings. Ask: what does “it” mean in each ending?
2. Who is “it”? (12 min): Write three sentences, such as “The goat ate the maize because it was hungry.” In pairs, pupils copy them and draw arrows from “it” to the words that tell them what it means. Compare the arrows on the board.
3. Change one word (6 min): Change “hungry” to “sweet”. Pupils redraw the arrows: now “it” is the maize. Ask: which single word changed the meaning?
4. The loop (9 min): Draw the loop: tokens, numbers, attention, scores, pick, repeat. Explain that attention does what the arrows did: the newest word weighs the earlier ones. Play “human language model”: the class scores three possible next words for a sentence and picks one.
5. Wrap-up (8 min): Show a chatbot answer with a wrong local fact (prepared in advance) and a textbook page. Ask: how would grounding help, and what would we still check? Then give the exit ticket.

Prepare three “it” sentences in the pupils’ language, a printed chatbot answer that contains one wrong local fact, and the textbook page that corrects it.

Discussion

- How did you know what “it” meant? Which words did you look back at?
- Why might a chatbot get a fact about our town wrong?
- What source would you give a chatbot before trusting its answer?

Exit ticket

- Write the loop in five words, and one reason a language model can be confidently wrong.

Try it in class tomorrow
Play “Who is ‘it’?”: write three sentences on the board where “it” could mean two things, such as “The goat ate the maize because it was hungry.” Pupils draw arrows from “it” to the words that tell

them what it means. Then change one word (“hungry” to “sweet”) and redraw the arrows. Explain that a language model’s attention does the same job: it weighs earlier words before it writes the next one.

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **Why can a language model sound sure and still be wrong?**

- ◯ It copied a wrong website
- ◯ It is lying on purpose
- ◯ It predicts likely text, not facts
- ◯ It skipped the fine-tuning stage

2 **Put one round of writing in order.**

- ☐ Pick one and add it to the text
- ☐ Score every possible next token
- ☐ Layers of neurons work on the numbers
- ☐ Attention: the newest token weighs the earlier ones
- ☐ Turn the tokens so far into lists of numbers

3 **If raters always prefer a full answer to "I don't know", fine-tuning teaches the model to guess when it doesn't know.**

- ◯ True
- ◯ False

4 **Match each stage to what happens in it.**

Pretraining	• • • •	Give it a trusted source to answer from
Fine-tuning with examples	• • • •	People rate which of two answers is better
Human feedback	• • • •	Learn from good answers that people wrote
Grounding	• • • •	Predict the next token across huge amounts of text

✎ **Where would grounding help most in your teaching? Which source would you give a chatbot, and what would you still check yourself?**